

POTHOLELOG · REQUIREMENTS

PotholeLog requirements

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DOCUMENT CONTROL

Document ID	PHL-REQ-001	Version	0.6
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/potholelog		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.6	2026-10-02	Amish Chadha	Decisions of 2026-10-02 (PHL-DEC-001): R13 restated for published dates (pooled, never pass times); R14 CityTwin ingest path decided; R15 notes the cost of the adopted light pipes and vent. No status changed
0.5	2026-10-01	Amish Chadha	Constructable design (PHL-DDR-003): R12 plate mode and R15 cost updated; budget treated as a value-engineering target
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 status from PHL-CAL-001; R16 redefined per PHL-DDR-001 (D1); R8, R13 and R14 rest on adopted choices
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-25	Amish Chadha	Initial scaffold

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These requirements are checked by calculation in PHL-CAL-001 v0.3. Targets are unchanged from v0.2 except R16, redefined for this build under PHL-DDR-001 (D1), and R5, whose defect-detection range is restated as 10 to 30 km/h under PHL-DDR-002 (N2). Both were decided by Amish on 2026-09-25 (go with recommendation). The status column gives the result on paper: "met on paper" means a calculation shows the target is met under the stated assumptions; "met by design" means the chosen parts and architecture meet it; "not verifiable at TRL 3" means only data from a vehicle can show it.

Table 1. Requirements.

ID	Requirement	Target	Status at TRL 3 (PHL-CAL-001)	Verification (later)
R1	Detect potholes and severe defects in the wheel path	80 % or more of defects at least 50 mm deep and 300 mm long found within 3 passes	At risk: a bus tyre drops only 22 mm into a 300 mm pothole; detectable up to about 20 km/h on IRI 4 roads and 50 km/h on IRI 2 roads	Detection model on logged data against a surveyed defect list
R2	Keep false reports low	10 % or fewer of reported defect clusters (2 or more passes) are not real defects	Not verifiable at TRL 3; Pothole Patrol reported over 90 % real after clustering	Field check of reported clusters
R3	Report roughness per road segment	One roughness value per 100 m segment per pass, calibrated per vehicle to an IRI estimate; r^2 of 0.8 or better against reference IRI on calibration sections	Not verifiable at TRL 3; synthetic calibration gives r^2 0.93 from one pass and 0.99 from five	Calibration runs following World Bank Technical Paper 46
R4	Locate defects	Defect cluster within 10 m of its true position (95 %) and on the correct street	At risk: 2.3 m suburban, 18.2 m in dense urban streets (10-pass cluster)	GNSS logs against surveyed points
R5	Work across normal fleet speeds	Roughness valid from 20 to 80 km/h; defect detection from 10 to 30 km/h, relying on repeat slow passes near stops and junctions (restated under PHL-DDR-002, N2; was 10 to 80 km/h)	At risk: margin 1.94 or more on IRI 2 roads across the band, but 0.97 at 30 km/h on IRI 4 roads (1.17 allowing 10 background crossings per km); roughness signal is 12 times sensor noise at 20 km/h	Speed-banded calibration
R6	Sample fast enough	3-axis acceleration at 400 Hz or more, range ± 8 g or more; GNSS position at 5 Hz or more	Met by design (400 Hz, ± 16 g, 10 Hz GNSS with PPS; floor peak 0.14 g)	Datasheets and firmware configuration
R7	Store data on the vehicle	30 days or more of raw data and summaries without upload	Met on paper (133 to 160 days on 32 GB)	Storage calculation
R8	Get data off the vehicle	Summaries uploaded automatically within 24 h, with no driver or staff action	Met on paper for vehicles in daily service (112 kB in about 8 s over depot Wi-Fi, with the ignition on)	Design review
R9	Run from vehicle power	9 to 36 V DC input (12 V and 24 V systems); reverse-polarity protected; transient levels of ISO 16750-2 as a target; under 1 W running; no draw with ignition off At risk: 60 V-rated converter (PHL-DDR-002, N1) clears the 58 V suppressed load dump and the 58.1 V TVS clamp by 1.9 V; TVS pulse energy during a load dump needs a bench test; running power 0.54 W met	Power budget; later bench test	
R10	Shut down cleanly	Files closed without corruption when power is cut at any time	Met on paper (7.0 s hold-up at end of life; 1 s needed)	Hold-up calculation

ID	Requirement	Target	Status at TRL 3 (PHL-CAL-001)	Verification (later)
R11	Survive the vehicle environment	Operating -20 to +70 °C; enclosure IP65; no loosening under vehicle vibration	At risk: about 72 °C inside at +70 °C needs 85 °C part grades; vibration not verifiable	Component ratings; later vibration test
R12	Install quickly and safely	30 min or less by a fleet mechanic; no cutting or welding of structural members; bracket first natural frequency above 150 Hz	At risk: 30 min task estimate at the limit; plate first mode 168 Hz met on paper (constructable design, PHL-DDR-003)	Design review
R13	Protect privacy	No camera or microphone; published data limited to road-segment roughness and defect locations, with no vehicle tracks, timestamps or driver identity; no driver scoring. A date per segment may be published, never pass times, and only once at least two vehicles or several days are pooled (PHL-DEC-001, 2026-10-02)	Met by design (privacy rule D7)	Design review; data schema review
R14	Open outputs	CSV and GeoJSON outputs that CityTwin and common GIS tools can read	Met by design (format D6); CityTwin fetches the segment files from the operator's server (PHL-DEC-001, 2026-10-02)	Sample export
R15	Low cost and buildable	Parts cost against the \$75 value-engineering target (a hypothetical control target, raised from \$70 under PHL-DDR-002, N1); off-the-shelf modules; no custom PCB	Met on paper (\$74.00, \$1.00 under the target)	Priced BOM
R16	Fit the fleet types in this build	Buses and refuse trucks; a bicycle variant is deferred to a later step (PHL-DDR-001, D1)	Met by design for buses and refuse trucks; bicycles are not served by this build	Design review

■ Requirements not met or at risk

- **R5 at risk.** The R1 reference pothole is largely bridged by a bus tyre, so its floor signal falls below ordinary road vibration at 30 km/h and above on fair roads (PHL-CAL-001, section C). Under PHL-DDR-002 (N2) the defect-detection range is restated as 10 to 30 km/h, with repeat slow passes near stops and junctions; the 300 mm reference defect is kept. The restated range is met on good roads and on fair roads up to about 29 km/h; at 30 km/h on an IRI 4 road the margin is 0.97. Longer potholes (600 mm) are detectable at all speeds.
- **R9 at risk.** Under PHL-DDR-002 (N1) the converter is now rated 60 V, which clears the 24 V suppressed load dump (58 V) and the TVS clamp (58.1 V), a margin of only 1.9 V. The TVS conducts during the load dump, and whether it survives the pulse energy can only be shown by a bench test (TRL 4, on hold).
- **R1 at risk.** Detection depends on passes at low speed; only field data can show whether 80 % within 3 passes is reached.
- **R4 at risk** in dense urban streets, where GNSS errors repeat between passes.
- **R11 and R12 at risk.** Part temperature grades, vibration loosening and the 30 min installation are all at their limits.
- **R2 and R3 not verifiable at TRL 3.** Both need field data and a calibration partner with reference IRI sections.

- **R15.** Value-engineering target: USD 75. Estimated cost of the constructable design: USD 74.00 (USD 1.00 under the target), after the 60 V converter (PHL-DDR-002) and the carrier plate and box fixing kit added for construction (PHL-DDR-003). A cellular option would take it over the target. The status light pipes and breather vent adopted on 2026-10-02 (PHL-DEC-001) are not yet in the BOM; together they would take the estimate about USD 2 to 3 over the target.
- **Bicycles.** The pitch still names bikes, but this build serves buses and refuse trucks only (R16 as redefined).

■ Assumptions

- Host vehicles run about 200 km a day (estimate for an urban bus or refuse truck), return to a depot with Wi-Fi each night and spend at least about 10 s with the ignition on within its coverage.
- The logger is mounted on the sprung mass (floor or body structure), not the axle; the suspension filters the input, which calibration corrects.
- 100 m segments match common practice for network roughness reporting; defects are reported as point events.
- A partner agency can provide reference IRI on a few calibration sections, or allow a rod-and-level survey of them.